

Clean Immediate-Opt-Out Baseline

A Formal Laboratory for Informational Power Through Pivotality

Working formal appendix

2026-08-04

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1 Status and scope

This document is the autonomous formal laboratory for Goal 1. It replaces the superseded delayed-continuation derivation with a clean benchmark in which

$$\pi_H = 0, \quad b_0 = b_1 = 0,$$

and a no vote by type θ of the hegemon is an immediate, irreversible opt-out. In Round 1 that action pays o_θ immediately and without discounting. It does not also give the hegemon a discounted continuation option.

The current version contains the complete Gate 0 contract. Analytical equilibrium results are not presumed below. They will be added only after an independent read-only audit gives Gate 0 a PASS without substantive reservation.

Current formal status: Gate 0 candidate; independent verdict pending.

Manuscript boundary: `formal_model_v6.Rmd` and `formal_model_v5.Rmd` are reference objects only. This Goal does not edit either manuscript.

Excluded from the baseline: $y + b_\theta$, a rejection payoff βC_θ chosen by a no vote in Round 1, the hybrid $\max\{o_\theta, \beta C_\theta\}$, the decomposition $t_\theta = d_\theta - b_\theta$, positive recognition probability for H , endogenous rule choice, sequential roll-call voting, and the archived feasibility/C-B-R architecture. A discounted Round-2 payoff after H votes yes and weak votes make the Round-1 ballot fail is part of the baseline; it is not the payoff from opting out.

1.1 Provenance

The pre-reset state is commit 84a644128586d4f4d81c022cd7d3e09454ee8004, recorded in the user-approved annotated tag pre-clean-optout-goal1-2026-08-03. The initial SHA-256 of formal_model_v6.Rmd is f18a999300c88e32bdc7542f3249cd258afd19a0c251c0b29c2c387a7131dbc1.

2 Gate 0: complete game contract

2.1 Players, type, prior, and parameters

There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states. Nature draws

$$\theta \in \{0, 1\}$$

before Round 1. The type persists. Only H observes θ . Weak states share the prior

$$\mu = \Pr(\theta = 1) \in [0, 1].$$

The Bayesian derivations use $\mu \in (0, 1)$. The endpoints are treated as degenerate-belief cases or limits, and not as positive-probability separating histories containing both types.

The weak coalition has a fixed, known institutional surplus normalized to one. The private type does not change this pie. Weak outside payoffs are zero. The clean primitive domain is

$$0 \leq o_0 < o_1 \leq \bar{y} \leq 1, \quad \beta \in (0, 1]. \quad (\text{G0-domain})$$

If a current agreement includes H , its payoff is y . Thus $b_0 = b_1 = 0$. The outside payoff o_θ is external to the institutional pie. It is never subtracted from weak surplus and is never paid together with y in the same history.

The institutional rule $R \in \{U, M\}$ is fixed exogenously for each comparison. There is no endogenous rule-choice stage.

2.2 Horizon, recognition, and time units

The game has exactly two bargaining rounds, R1 and R2. There is no R3. In each round that is reached, a new proposer is drawn publicly and independently of the previous recognition, the type, and the public history. The baseline fixes

$$\pi_H = 0$$

in every round. Each weak state is recognized with probability $1/m$, conditional on that round being reached.

Round 1 defines the payoff unit. A payoff realized in R2 is multiplied by β in Round-1 units. A no vote by H in R1 pays o_θ at once, without β . A no vote by H in R2 pays o_θ at the R2 date, or βo_θ in Round-1 units.

R2 is terminal. If H is still active and its R2 ballot fails, then H receives o_θ at the R2 date, regardless of whether it voted yes or no; weak states receive zero. This is the terminal outside payoff, not a second payment. If H opted out in R1, it receives o_θ exactly once and receives no R2 flow.

2.3 Formation

After Nature draws θ , but before the R1 recognition draw, the m weak states make one collective, all-or-nothing formation decision. They do not observe θ . The hegemon observes its type but takes no formation action.

If the institution forms, every weak state pays an external sunk cost $\chi \geq 0$. The cost does not reduce the unit pie. If the institution does not form, every weak state receives zero and H receives o_θ .

For rule R , the gross payoff relevant to collective formation is

$$V_W^R(\mu) = \frac{1}{m} \mathbb{E} \left[\sum_{j=1}^m u_j^R \middle| \text{formation} \right]. \quad (\text{G0-entry-value})$$

The expectation integrates over the type, recognition, public histories, and any symmetric selection among weak coalition partners. Formation occurs iff $V_W^R(\mu) - \chi \geq 0$; equality leads to formation. This collective primitive is not an assertion that every realized weak state individually prefers formation under an asymmetric coalition selection.

2.4 Contract space and budget identities

Suppose weak state i is proposer. While H is active, a proposal is

$$s_i = (y, (x_j)_{j \neq i}),$$

where

$$y \in [0, \bar{y}], \quad x_j \geq 0,$$

and the proposer's residual if H participates is

$$x_i^H = 1 - y - \sum_{j \neq i} x_j \geq 0. \quad (\text{G0-H-budget})$$

The same space applies in R1 and R2 and under both rules. The component y is a public offer conditional on the observed participation of H ; there is no hidden inclusion indicator. If H votes yes and the proposal passes, it is included and receives y . Every named weak-state offer x_j is paid when the proposal passes, independently of that recipient's individual vote.

If H votes no under majority and the weak votes nevertheless reach the quota, the agreement passes without H . The conditional component y is not burned and is not paid to H . It is reabsorbed by the weak proposer, so

$$x_i^{WO} = 1 - \sum_{j \neq i} x_j = x_i^H + y. \quad (\text{G0-WO-budget})$$

The hegemon receives o_θ externally. Hence both implemented budget identities close at one:

$$y + \sum_{j \neq i} x_j + x_i^H = 1, \quad \sum_{j \neq i} x_j + x_i^{WO} = 1. \quad (\text{G0-budget-closure})$$

After an earlier opt-out, only the m weak states remain active, every proposal fixes $y = 0$, and the weak-only identity in (G0-WO-budget) applies. If a ballot fails, no current x_j and no current y is implemented.

These rules make y contingent only on publicly observed participation, not on the private type.

2.5 Common simultaneous ballot

The game is sequential and public across rounds: proposal, ballot, publication of the complete vote vector and outcome, followed by any continuation. It is not sequential within a ballot.

The proposer is counted as voting yes. All other active players observe the proposal and public history and vote simultaneously. The individual votes are revealed only after every ballot has been cast. The same ballot protocol is used under both rules. Only the quota changes:

$$Q_U = N, \quad Q_M = q = \left\lfloor \frac{N}{2} \right\rfloor + 1. \quad (\text{G0-quotas})$$

Because $N \geq 3$, $q \leq m$. While H remains active, the electorate contains all N states. After opt-out, the active electorate contains only the m weak states, but the majority quota remains the original q ; it is not recalculated. The unanimity quota remains N . Therefore an opt-out makes future unanimity impossible and that branch ends immediately.

Players vote yes when indifferent. Strategies are functions of information available before the ballot closes. No strategy may condition on a simultaneous vote that has not yet been observed.

2.6 Immediate opt-out

A no vote by type θ of H is an immediate and irreversible opt-out. No current or future agreement can then include H . In R1 the action pays o_θ immediately. In R2 it pays o_θ at the R2 date. There is no additional choice between opt-out and a discounted continuation.

Under unanimity, an R1 no vote by H makes the quota impossible and ends the branch. Under majority, if the weak yes votes already reach q , their current weak-only agreement passes. Otherwise the game proceeds to an R2 weak-only subgame with the original quota q . A joint R1 rejection by H and weak voters follows the same rules.

A yes vote by H is not an opt-out. If the R1 ballot then fails only because weak votes do not reach the quota, H remains active and the game proceeds to R2 under the same institutional rule.

2.7 Exhaustive voting-history table

Let z denote the number of weak yes votes, including the weak proposer's automatic yes vote. Table 1 is generated from the versioned tab-separated source `tables/clean_optout_gate0_histories_piH0.tsv`. The concise entries use `C_H2^R(theta,nu)` only for the legitimate continuation after an R1 yes vote by H followed by a weak-caused failure. They never use that object as the payoff from a no vote.

Table 1: Gate 0 exhaustive voting histories, implementation rules, payoffs

history_id	round	rule	active players	quota	quota reached?	H vote	participation offer to H?	failure cause	H included?	H remains?	y implemented?	destination of y	H payoff
G01	R1	U	H plus m weak	N	yes: $z=m$	yes	yes	none	yes	no continuation	yes	paid to H	y
G02	R1	U	H plus m weak	N	no: $z < m$	yes	yes	weak-vote shortfall	no	yes	no	not implemented	$\beta \theta C$
G03	R1	U	H plus m weak	N	no: $z=m$	no	yes	H-no is decisive	no	no	no	not implemented	θ
G04	R1	U	H plus m weak	N	no: $z < m$	no	yes	joint H-no and weak rejection	no	no	no	not implemented	θ
G05	R1	M	H plus m weak	q	yes: $z \geq q-1$	yes	yes	none	yes	no continuation	yes	paid to H	y
G06	R1	M	H plus m weak	q	no: $z \leq q-2$	yes	yes	weak-vote shortfall	no	yes	no	not implemented	$\beta \theta C$
G07	R1	M	H plus m weak	q	yes: $z \geq q$	no	yes	none; weak votes reach q	no	no	no	reabsorbed by proposer	θ
G08	R1	M	H plus m weak	q	no: $z \leq q-1$	no	yes	H-no is decisive for failure	no	no	no	not implemented	θ
G09	R1	M	H plus m weak	q	no: $z \leq q-2$	no	yes	joint H-no and weak shortfall	no	no	no	not implemented	θ
G10	R2	U	H plus m weak	N	yes: $z=m$	yes	yes	none	yes	no continuation	yes	paid to H	$\beta \theta y$
G11	R2	U	H plus m weak	N	no: $z < m$	yes	yes	weak-vote shortfall	no	no	no	not implemented	$\beta \theta$
G12	R2	U	H plus m weak	N	no: $z=m$	no	yes	H-no is decisive	no	no	no	not implemented	$\beta \theta$
G13	R2	U	H plus m weak	N	no: $z < m$	no	yes	joint H-no and weak rejection	no	no	no	not implemented	$\beta \theta$
G14	R2	M	H plus m weak	q	yes: $z \geq q-1$	yes	yes	none	yes	no continuation	yes	paid to H	$\beta \theta y$
G15	R2	M	H plus m weak	q	no: $z \leq q-2$	yes	yes	weak-vote shortfall	no	no	no	not implemented	$\beta \theta$
G16	R2	M	H plus m weak	q	yes: $z \geq q$	no	yes	none; weak votes reach q	no	no	no	reabsorbed by proposer	$\beta \theta$
G17	R2	M	H plus m weak	q	no: $z \leq q-1$	no	yes	H-no is decisive for failure	no	no	no	not implemented	$\beta \theta$
G18	R2	M	H plus m weak	q	no: $z \leq q-2$	no	yes	joint H-no and weak shortfall	no	no	no	not implemented	$\beta \theta$
G19	R2	M	m weak; H already opted out	original q	yes: $z \geq q$	not applicable	no; y fixed at 0	none	no	no	no; fixed at 0	not available	0 addition
G20	R2	M	m weak; H already opted out	original q	no: $z < q$	not applicable	no; y fixed at 0	weak-vote shortfall	no	no	no; fixed at 0	not available	0 addition
G21	R2	U	m weak; H already opted out	N	structurally impossible	not applicable	no; y fixed at 0	H absent, so unanimity quota cannot be met	no	no	no; fixed at 0	not available	0 addition

The R2-unanimity row with H already absent is included to document a structurally unreachable class: the branch ends when the opt-out occurs because the original quota N cannot be met by m active weak states. It is not a latent R2 action.

3 Information, beliefs, and incentive constraints

3.1 Public histories and beliefs

The public history after a ballot contains the proposal, complete vote vector, quota outcome, and any observed opt-out. On-path beliefs follow Bayes' rule.

The baseline uses the expression **weak-vote-passive assessment**. It is a maintained assessment of the simultaneous public ballot, not a refinement, D1, the Intuitive Criterion, a sequential-equilibrium restriction, or a characterization of all PBEs.

A unilateral vote deviation by a weak state is not treated as direct evidence about θ , because weak states do not observe θ . Following a weak-voter deviation, the posterior is the posterior implied by the public proposal and H 's simultaneous type-contingent strategy and vote, not by the action of an uninformed deviator. A separating vote by H can update beliefs when the continuation keeps H active and the type remains payoff-relevant.

Every zero-probability continuation in which H remains active must state the required off-path posterior and identify which result depends on it. Off-path posteriors are components of an assessment, not primitives of the game. They cannot be selected to delete an inconvenient branch. Once H opts out, θ is payoff-irrelevant for the weak-only continuation and no posterior is needed to solve it.

3.2 Hegemon's expected ballot IC

Fix an R1 public history h , a proposal, and type θ . Let v_W be the simultaneous vector of non-proposer weak votes, let $A_R^Y(v_W)$ indicate that the rule- R quota is reached when H votes yes, and let $\sigma_W(v_W | h)$ be the probability induced by the weak strategies. The correct R1 payoff from voting yes is

$$EU_H^R(Y | \theta, h) = \sum_{v_W} \sigma_W(v_W | h) [A_R^Y(v_W)y + \{1 - A_R^Y(v_W)\}\beta C_{H,2}^R(\theta, \nu(h, Y, v_W))]. \quad (\text{G0-H-yes-R1})$$

The payoff from voting no is always

$$EU_H^R(N | \theta, h) = o_\theta. \quad (\text{G0-H-no-R1})$$

This equality holds whether a majority weak-only agreement passes immediately or the weak states continue alone: H has opted out and receives no further flow. Sequential rationality requires

$$EU_H^R(Y | \theta, h) \geq o_\theta \quad (\text{G0-H-IC-R1})$$

for a prescribed yes vote, with the reverse weak inequality for a prescribed no vote.

At an R2 information set, a failed ballot with H still active pays the terminal outside payoff. Thus, in Round-1 units,

$$EU_H^R(Y | \theta, h_2) = \beta \sum_{v_W} \sigma_W(v_W | h_2) [A_R^Y(v_W)y + \{1 - A_R^Y(v_W)\}o_\theta], \quad EU_H^R(N | \theta, h_2) = \beta o_\theta. \quad (\text{G0-H-IC-R2})$$

Only when the weak strategies make $A_R^Y(v_W) = 1$ with probability one does the R1 or R2 yes/no comparison reduce to

$$y \geq o_\theta. \quad (\text{G0-direct-cutoff})$$

Therefore the direct threshold $y_\theta^* = o_\theta$ is derived at a certain-implementation information set. It is not a global voting rule. If weak-caused failure has positive probability, equation (G0-H-IC-R1) must integrate current implementation and the legitimate R2 continuation after a yes vote.

3.3 Weak-voter approval IC

Fix a non-proposing weak voter j . Its strategy is chosen before the simultaneous ballots are revealed. Let $u_j(a_j, v_{-j})$ be the payoff specified by the history table: an implemented x_j if the current agreement passes, the appropriate discounted continuation if R1 fails, or zero after terminal failure. Its approval condition is

$$\sum_{v_{-j}} \Pr(v_{-j} \mid h) u_j(Y, v_{-j}) \geq \sum_{v_{-j}} \Pr(v_{-j} \mid h) u_j(N, v_{-j}). \quad (\text{G0-W-IC})$$

The expectation includes H 's type-contingent simultaneous vote. A voter cannot condition on that vote before the ballot closes. If voter j is pivotal with probability one, equation (G0-W-IC) reduces to the usual comparison between x_j and the continuation induced by a no vote. If its vote never changes implementation or continuation, it is indifferent and votes yes. Offers are implemented for named recipients whenever the agreement passes, even if the recipient voted no.

3.4 Tie-breaking

Every player votes yes when indifferent. Among proposals that maximize the weak proposer's payoff, the selected proposal minimizes H 's expected payoff. No third tie-break is available. If two proposals tie for both the weak proposer and H but imply different downstream objects, the result must remain set-valued or exclude that boundary by a stated strict theorem domain; the proof may not select one silently.

4 Gate 0 audit checklist

The implementer records the following mechanical assessment, which does not substitute for the independent verdict.

Requirement	Candidate evidence	Implementer status
Every required history has a payoff and continuation	Table 1, rows G01–G21	checked
No double payment to H	Contract identities and all H-no rows	checked
No reinclusion after opt-out	Rows G03, G04, G06, G08, G11, G13, G15–G18	checked
No discounted continuation as payoff from H-no in R1	Equation (G0-H-no-R1) and table H-no rows	checked
Same simultaneous ballot under U and M	Equation (G0-quotas)	checked
Post-opt-out electorate and quotas defined	Common-ballot section and rows G16–G18	checked
H-including and weak-only budgets close	Equations (G0-H-budget)–(G0-budget-closure)	checked
H IC integrates simultaneous weak votes	Equations (G0-H-yes-R1)–(G0-H-IC-R2)	checked
Beliefs stated whenever type remains relevant	Beliefs section and G02/G07	checked
No equilibrium result presumed as a primitive	Status, primitives, and proof ledger	checked

Independent Gate 0 verdict: PENDING.

5 Phase 1 placeholder

No equilibrium candidate, historical label, threshold ordering, No-Cheap-H condition, entry nesting result, or institutional comparison has yet been promoted under the clean contract. The next derivation must proceed from terminal histories backward only after Gate 0 receives an independent PASS.

5.1 Preliminary survival matrix

Historical result	Current status before rederivation	Required evidence
Terminal low-only/pooling choice	pending	derive from R2 direct opt-out thresholds
R1 pooling label P	historical candidate only	derive H and weak-voter ICs from Gate 0
R1 low-only label L	historical candidate only	derive H and weak-voter ICs from Gate 0
R1 rejection/delay label R or D	historical candidate only	exhaust immediate-opt-out deviations; do not impose delay
Rejected-history reduction lemma	historical only	reprove or reject under the new H-no payoff
No-Cheap-H	historical only	rederive with primitive o_θ
Majority no-screening	historical only	compare all payoff-relevant majority coalitions
Entry nesting	historical only	derive total weak payoffs under selected clean paths
H payoff comparison and classification	historical only	derive only after all preceding objects are proved

6 Proof ledger

Object	Status	Notes
Gate -1 provenance	proved	Clean HEAD, user-approved annotated tag, and initial v6 hash recorded.
Gate 0 primitives and timing	specified	Awaiting independent audit.
Gate 0 contract space and budgets	specified	Awaiting independent audit.
Gate 0 voting histories	specified and mechanically checked	Source is the versioned TSV; independent audit pending.
Gate 0 beliefs and IC definitions	specified	Expected IC explicitly integrates simultaneous weak votes.
R2 unanimity	pending	No claim promoted.
R2 majority with H active	pending	No claim promoted.
R2 majority weak-only	pending	No claim promoted.
R1 unanimity	pending	Historical P/L/R labels are not primitives.
R1 majority and No-Cheap-H	pending	Must be rederived under primitive outside payoffs.
Entry, nesting, and classification	pending	Must follow the equilibrium derivation.
Delayed continuation and other architectures	extension/history only	Not part of this Goal's baseline.